HW3

Based on today's class (Week 3), please complete the template below and include additional experimental results regarding backward, forward, and full schema compatibility modes.

Topic: Internet Service Provider Data Usage on Customers

All figures are in SVG format (where available).

1. Overview (2-3 sentences)

   Internet Service Providers will have TONS of data, I am focusing on the Data Usage aspect which we can use to analyze the customers behavior and introduce new packages/promotions that best fit the user in accordance to their behavior.

2. More specific details (2-3 sentences)

   I have inputted in customerID, their type of device (devicetype) whether it is tablet or phone or other types, their internet usage (bytes_used), and their time calling in minutes (minutes_called)

3. Existing issues (2-3 sentences)

   Currently only have standard data, no promotional code and current packages that they are currently using.

4. Motivation (2-3 sentences)

   To be able to handle lots of data, and have usable data to be analyzed for further uses.

5. Problem statement (Input and Objective)

   User Behavior Analytics. Input a new Promotional Code column for a one month trial (add in v2) and remove after one month (delete in v3) to test backward and forward compatibility.

6. Experimental results

   6.1 Backward mode

For the Backward mode, I went and add the “promotional_code” column with default being “null” (though the main type is “int”, I learned the we can string it together with [“null”, “int”] in “type” to have “null” as the default. Tested both promotional_code as 999 and None to test “int” type and “null” type

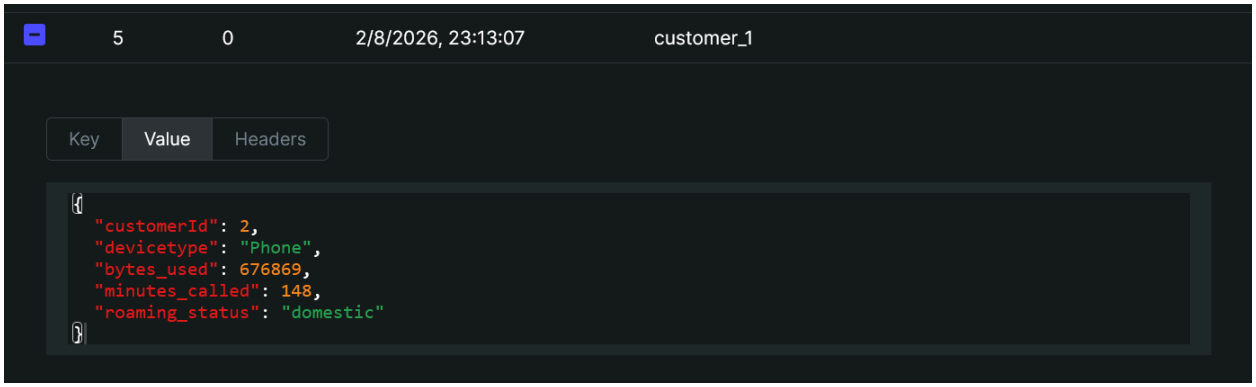
```
# 1. Define the Avro Schema
avro_schema_str = """
{
  "namespace": "example.avro",
  "type": "record",
  "name": "DataUsage",
  "fields": [
    {"name": "customerId", "type": "int"},
    {"name": "devicetype", "type": "string"},
    {"name": "bytes_used", "type": "long"},
    {"name": "minutes_called", "type": "int"},
    {"name": "promotional_code", "type": ["null", "int"], "default": null }
  ]
}
"""
```

6.2 Forward mode

The interesting thing for Forward Compatibility Mode is that Kafka is “smart”, as I input only “promotional_code” to be the only one that is added in v2 and will be deleting it in v3, v3 columns will look exactly the same as v1, so there is no new version, Kafka just use v1. (The new messages got sent but no new version was created because it is the same as v1.)

6.3 Full mode

Added roaming_status without Consumer crashing. Compatibility is set to “FULL”



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